

Ahmed Khalid Humoudi

Robotics and Automation Engineer

✉ ahmed_aku@hotmail.com ☎ +966 597166553 📍 Jeddah, Saudi Arabia 📅 12/12/2004

🌐 www.linkedin.com/in/ahmed-humoudi-904127317

PROFILE

Dedicated robotics engineering student with hands-on experience in control, automation, AI integration, and system development. Passionate and seeking opportunities to apply technical expertise to real-world applications.

EDUCATION

German International University (GIU), Bachelor Degree, 2026 09/2022 – Present | Cairo, Egypt

Major: Robotics and Automation

GPA: 0.87 (German System), equivalent to **3.85** on the United States 4.0 GPA scale.

Ranked among the top 3 students in the Robotics Major.

Global International School (Al-Kon), High School Degree, 2022 09/2019 – 06/2022 | Jeddah, Saudi Arabia
IGCSE Score: 120%

PROFESSIONAL EXPERIENCE

Mechatronics Engineer Intern, Hiryo 02/2026 – 07/2026 | Cairo, Egypt

Designed and developed a Bluetooth-based alarm and tracking device with real-time alert and location features, integrating embedded hardware, firmware, and mobile connectivity to deliver a reliable end-to-end system.

Physics Lab Assistant, German International University 02/2024 – 06/2024 | Cairo, Egypt

Conducted hands-on laboratory supervision and technical guidance, assisting in experimental procedures and the practical application of theoretical concepts.

PROJECTS

Bachelor Thesis

Reinforcement Learning-Based Path Planning for Warehouse Robots in Dynamic Environments:

Developed an RL-based navigation system for autonomous warehouse robots operating in dynamic environments with obstacle avoidance and adaptive path planning.

Multi-Robot Task Formation: Centralized ROS2 system, obstacle avoidance, path planning, formation control.

PLC-Based Stepper Motor Motion Control: controlling stepper motor speed, direction, position control.

UR5E Pick and Place: UR5e robot integrated with ROS 2, MoveIt 2, and RViz for trajectory planning.

Oven Control System: ATmega32, temperature monitoring, fan control, and safety lock system.

Ball and Beam Control System: PID-based ball stabilization on a beam using Arduino Uno.

4-DOF Robotic Arm: Arduino-based 4-DOF arm with inverse kinematics and trajectory planning.

TECHNICAL SKILLS

Programming: PLC, Embedded C, C++, JavaScript, Python, MATLAB, ROS, React Native

Design/Simulation: AutoCAD, SolidWorks, Fusion, Proteus, Simulide

Electronics: Arduino, Raspberry Pi, Atmega32, ESP32

Artificial Intelligence: Machine Learning, Deep Learning, Reinforcement Learning

Microsoft Office: Word, Excel, PowerPoint

Languages: Arabic (Native), English (C1), German (A2)

COURSES

Modern Robotics Foundations of Robot Motion, Northwestern University, Coursera Online

Introduction to Artificial Intelligence, IBM, Coursera Online

MATLAB Onramp, MATLAB Self-Paced Online Courses

SIMULINK Onramp, MATLAB Self-paced Online Courses

Reinforcement Learning Onramp, MATLAB Self-Paced Online Courses